

THEORY OF COMPUTATION

ASSIGNMENT 3

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1. For each language below,

- if the language is context free, provide a context-free grammar that generates it **and** a pushdown automaton that recognizes it;
- otherwise, prove that it is not context free.

$$1.1 L_1 = \{0^n 0^n 1^n \in \{0,1\}^* \mid n \geq 0\}$$

$$1.2 L_2 = \{0^n 1^n 0^n \in \{0,1\}^* \mid n \geq 0\}$$

$$1.3 L_3 = \{w \in \{0,1\}^* \mid \#0(w) = \#1(w)\}$$

where $\#0(w)$ denotes the number of occurrences of 0 in w , and similarly for $\#1(w)$.

2. Suppose G is the following grammar over the alphabet $\Sigma = \{a, b\}$:

$$S \rightarrow SAS \mid A$$

$$A \rightarrow aAb \mid B$$

$$B \rightarrow bBa \mid \epsilon$$

Find a grammar G' in Chomsky normal form that is equivalent to G .

3. Suppose G' is the grammar in Chomsky normal form that you obtained in the previous problem, apply Cocke-Younger-Kasami (CYK) algorithm to determine whether G' generates the string

$$w = baabab$$

or not. If so, also draw a parse tree of w in G' .